Haden

An AI-powered companion designed to support adopted children through personalized conversations and real-time emotion recognition

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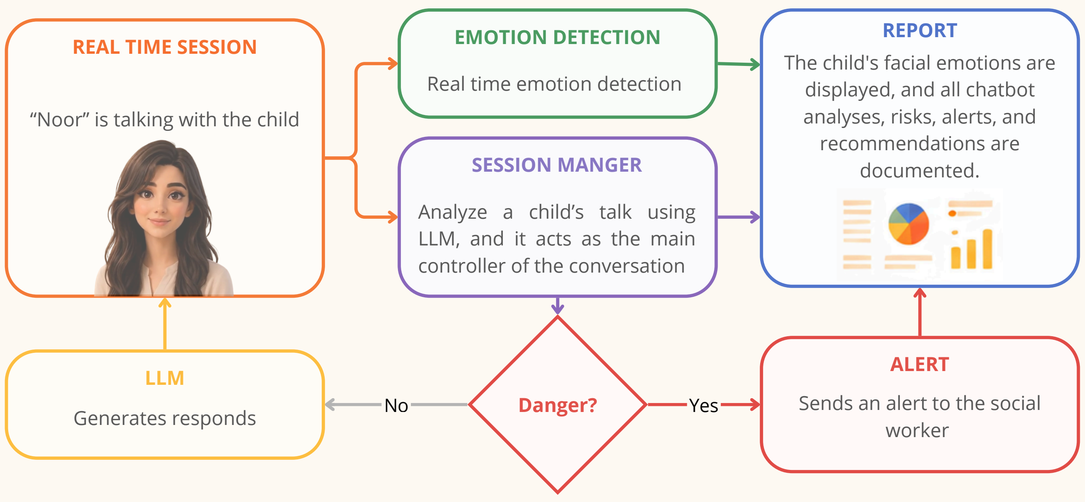
01 INTRODUCTION

Care institutions often rely on periodic visits and manual evaluations to monitor adopted children, which can delay the detection of psychological or behavioral issues. Therefore, the “Haden” project aims to develop an AI-based system that continuously monitors children and detects problems earlier by analyzing their emotions and behavior during interaction with a virtual companion.

02 OBJECTIVES

- Provide an intelligent system that supports social workers in monitoring adopted children.
- Detect risk indicators such as extreme sadness, fear, or abuse.
- Generate automated reports that support early intervention

03 SYSTEM ARCHITECTURE

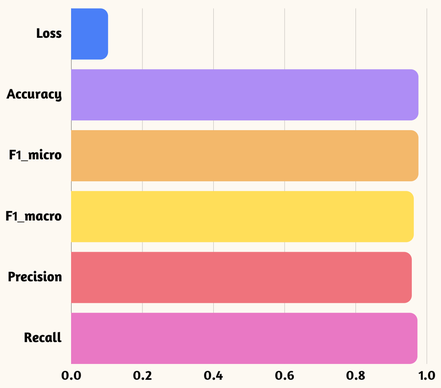


04 TOOLS

05 MODELS RESULTS

Face Emotion Detection



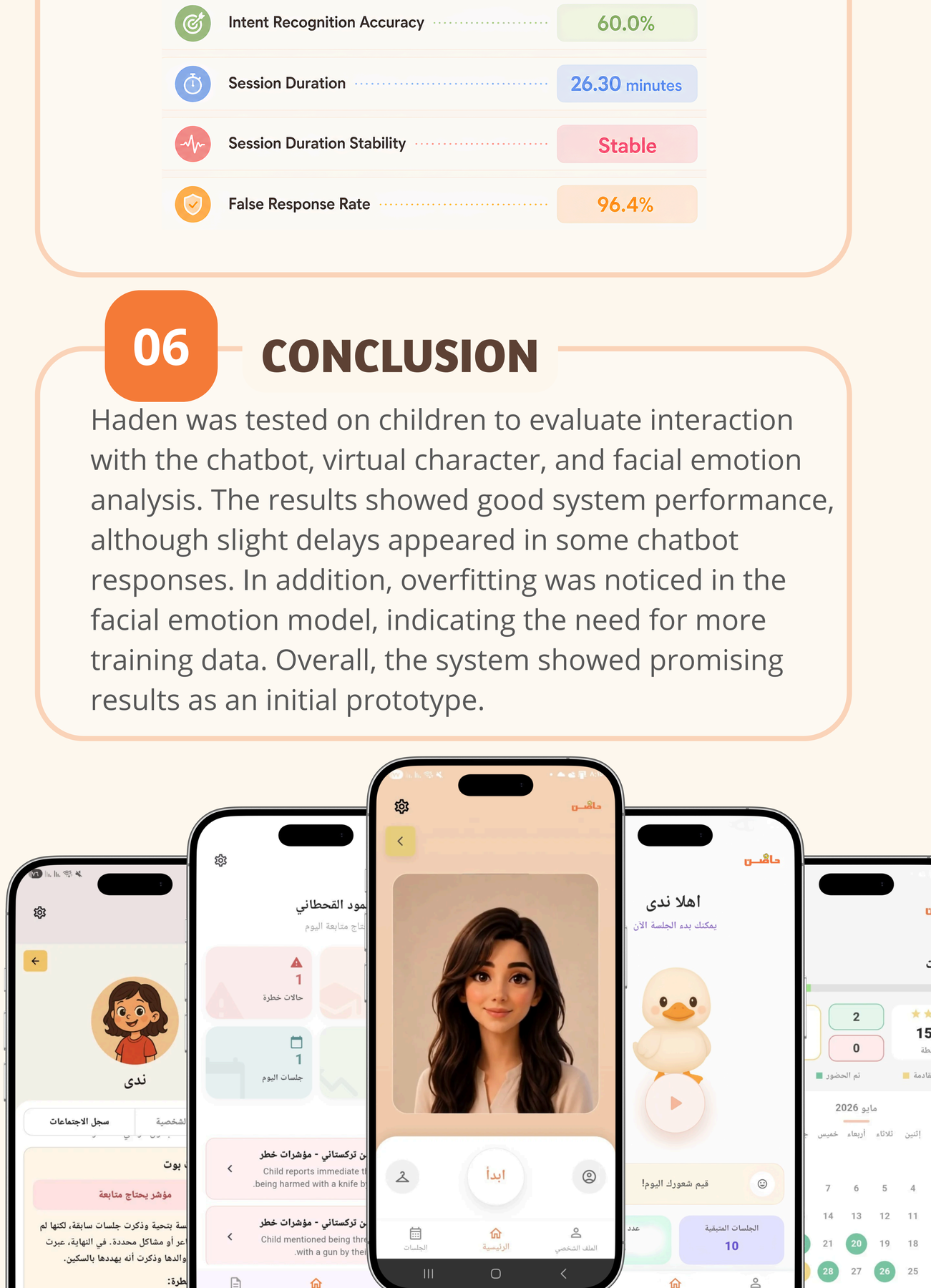


Chatbot Voice

Average Response Time	3.59 sec
Intent Recognition Accuracy	60.0%
Session Duration	26.30 minutes
Session Duration Stability	Stable
False Response Rate	96.4%

06 CONCLUSION

Haden was tested on children to evaluate interaction with the chatbot, virtual character, and facial emotion analysis. The results showed good system performance, although slight delays appeared in some chatbot responses. In addition, overfitting was noticed in the facial emotion model, indicating the need for more training data. Overall, the system showed promising results as an initial prototype.



FOR MORE INFORMATION